# Simulation Output Summary

Integrated Energy Systems Optimiser (IESO) v24.03 · www.github.com/greoux-research/ieso

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## Demand

Demand for Electricity:

- Annual Demand in MWh: 50000000
- Hourly Profile: Figure 1
- Non-Service Penalty in $ per MWh: 20000
- Supply Sources: Power Grid

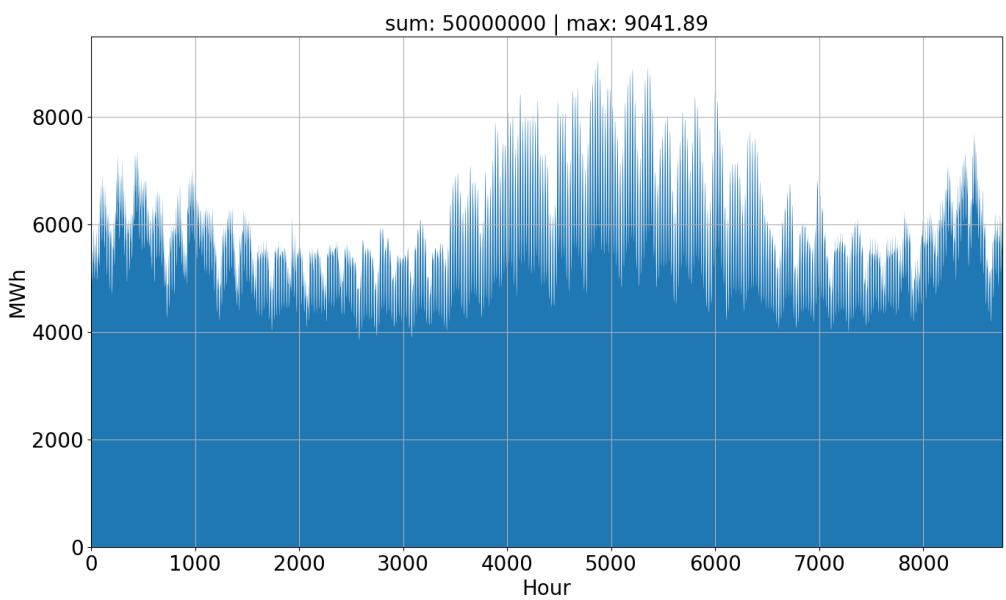


Figure 1: Demand for Electricity

## Generators

Generator "nucl":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 456466
- Variable Costs in $ per MWh: 11.5
- Variable Emissions in $ per MWh: 0
- The optimiser did not suggest adding generation capacity

Generator "coal":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 222392
- Variable Costs in $ per MWh: 26.84
- Variable Emissions in $ per MWh: 880
- Capacity in MW: 2265.43
- Electricity Generation: 12720644 MWh (Figure 2)
- Number of Successive Changes in Output: 2476

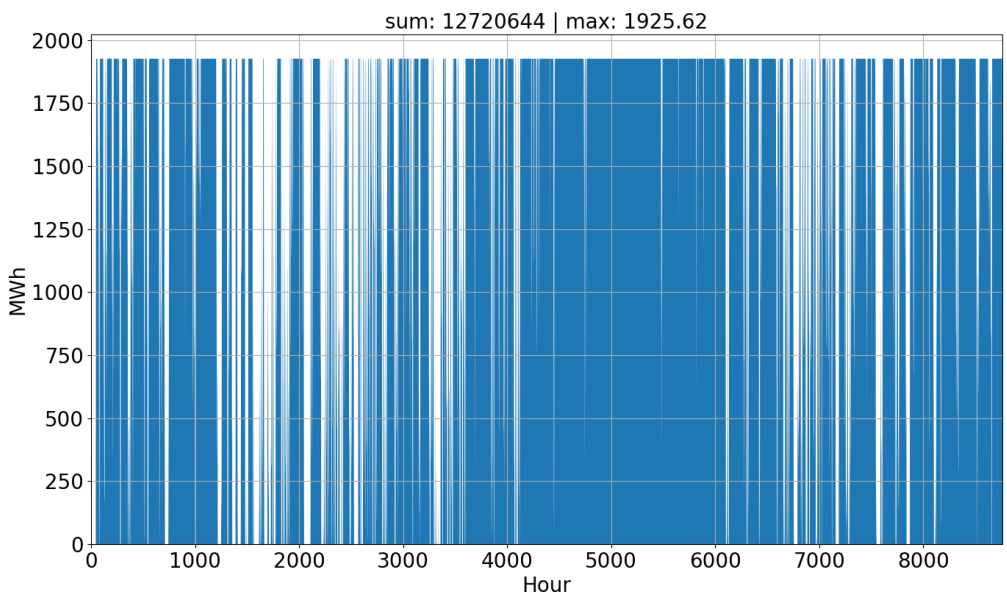


Figure 2: Electricity Generation by "coal"

Generator "ccgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 98749
- Variable Costs in $ per MWh: 56.44
- Variable Emissions in $ per MWh: 365
- Capacity in MW: 2936.85
- Electricity Generation: 6514881 MWh (Figure 3)

- Number of Successive Changes in Output: 3540

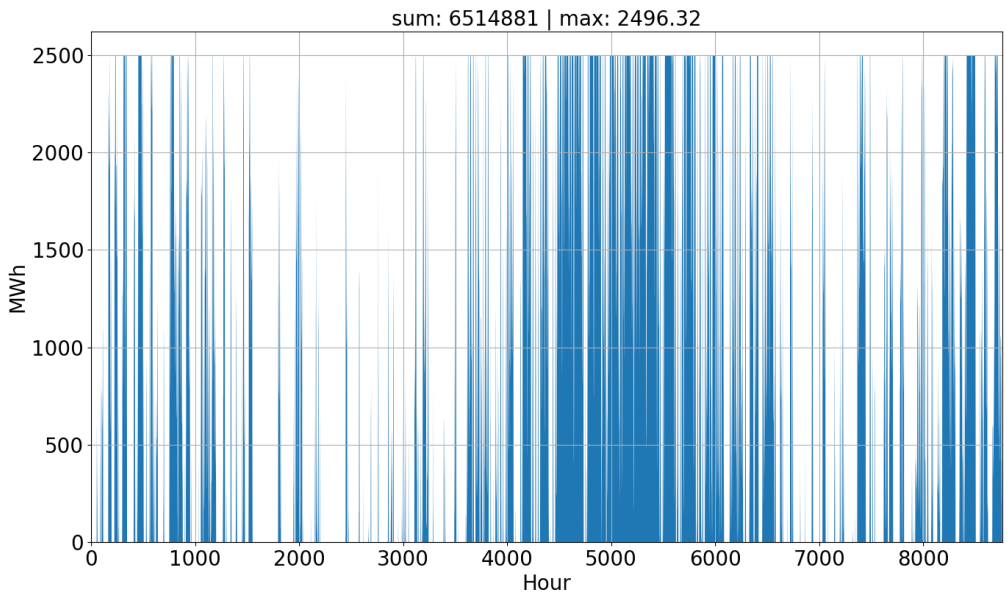


Figure 3: Electricity Generation by "ccgt"

Generator "ocgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 75182
- Variable Costs in $ per MWh: 96.11
- Variable Emissions in $ per MWh: 523
- Capacity in MW: 2228.3
- Electricity Generation: 364230 MWh (Figure 4)
- Number of Successive Changes in Output: 566

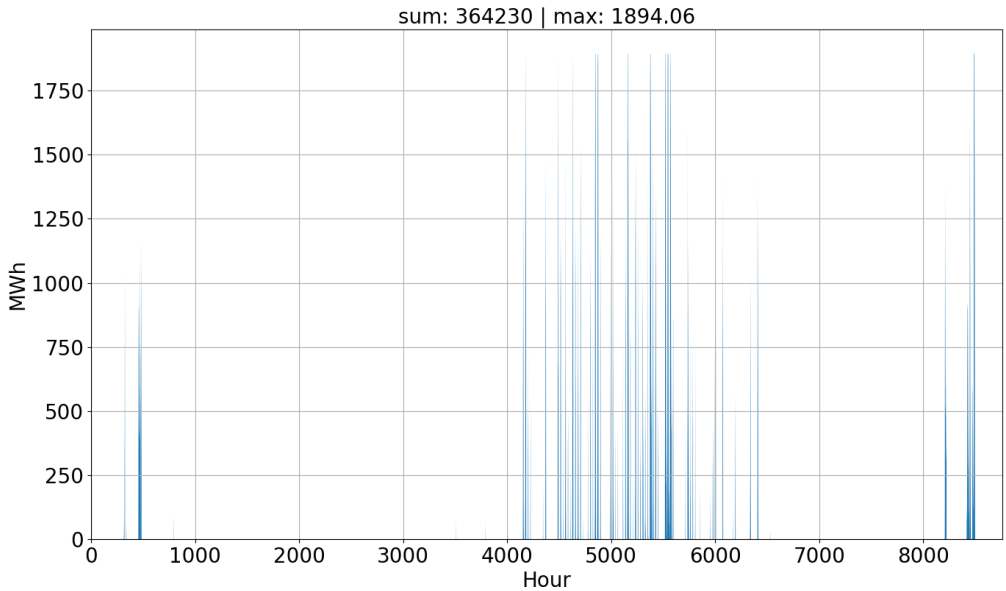


Figure 4: Electricity Generation by "ocgt"

Generator "solr":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 70046
- Variable Costs in $ per MWh: 3e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 4985.25
- Electricity Generation: 8195689 MWh (Figure 5)
- Curtailment: 0 Hours; 0 MWh or 0.0 % of the Tech.'s Annual Gen. Volume

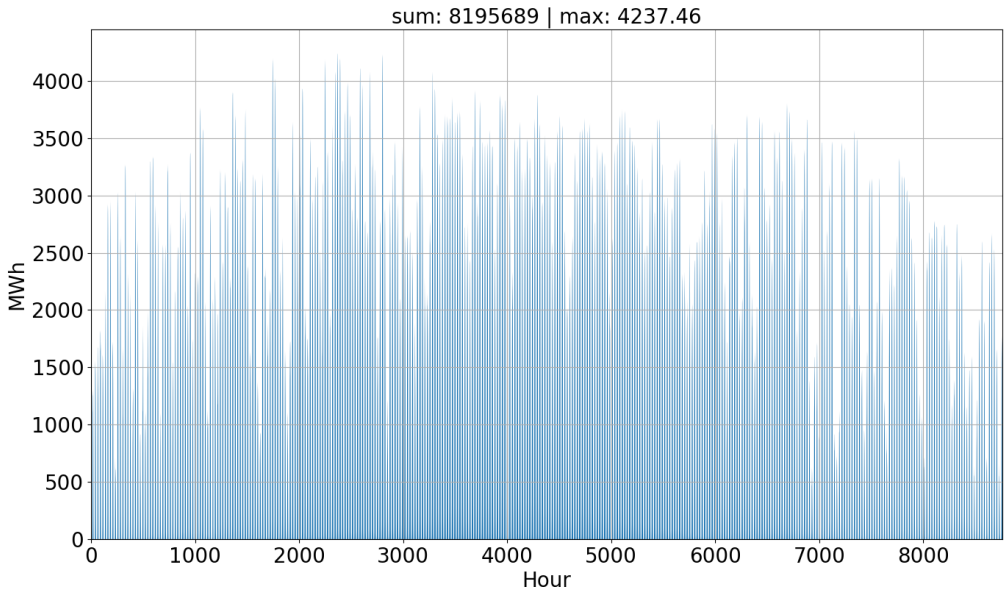


Figure 5: Electricity Generation by "solr"

Generator "wind":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 133553
- Variable Costs in $ per MWh: 6e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 6653.09
- Electricity Generation: 22336033 MWh (Figure 6)
- Curtailment: 565 Hours; 535354.84 MWh or 2.4 % of the Tech.'s Annual Gen. Volume

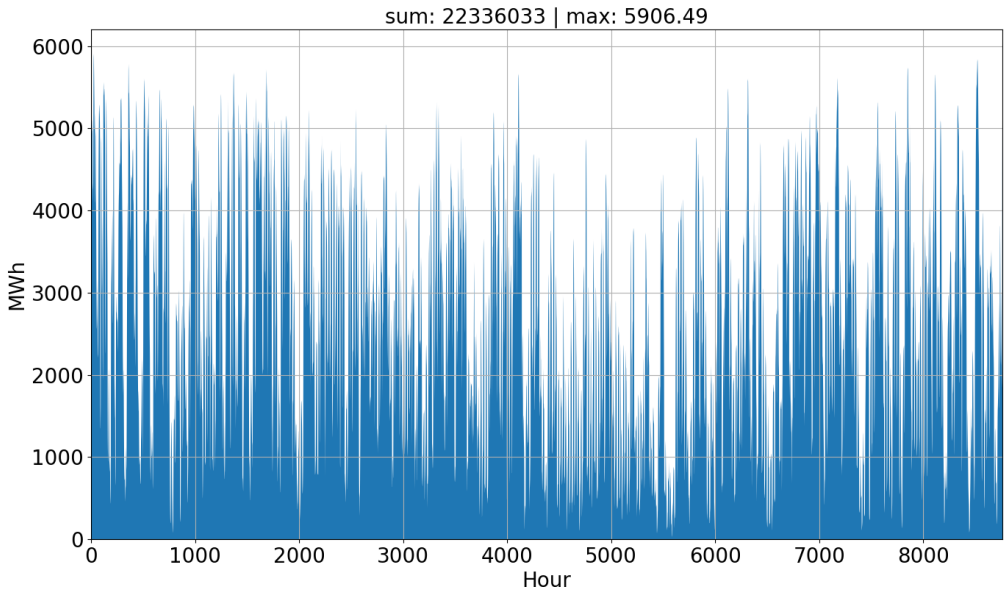


Figure 6: Electricity Generation by "wind"

## Flexibility Means

Flexibility Mean "bstr":

- Type: Electricity Storage System
- Fixed Costs in $ per MWh per Year: 27248
- Hours of Storage at Maximum Discharge: 4
- Round Trip Efficiency: 0.85
- Capacity in MWh: 3758.04
- Electricity Charged (-) and Discharged (+): Figure 7
- Electricity Stored: Figure 8

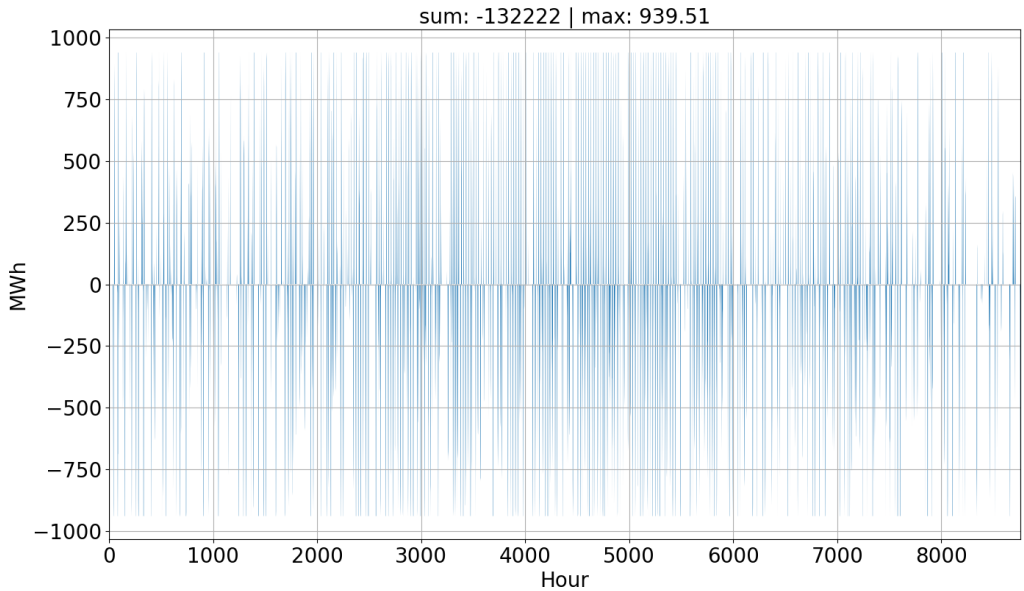


Figure 7: Electricity Charged (-) and Discharged (+) by "bstr"

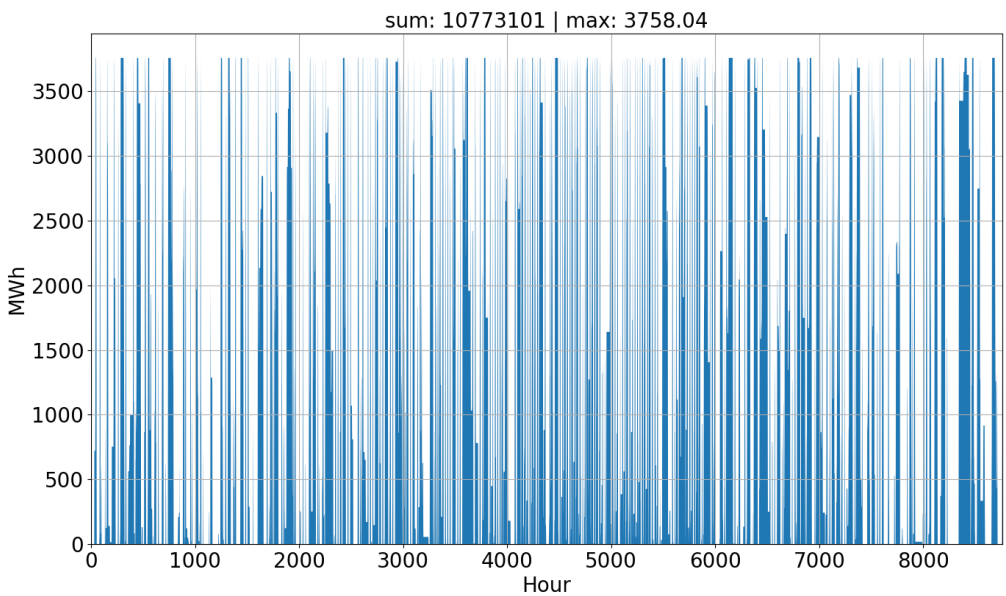


Figure 8: Electricity Stored by "bstr"

## Costs and Emissions

Generation Costs and Emissions on a per MWh Basis:

- Generation Costs in $ per MWh: 58.71
- Generation Emissions in kg per MWh: 274.53

## Prices

Electricity:

- Average Price in $ per MWh: 61.21
- Price Profile: Figure 9

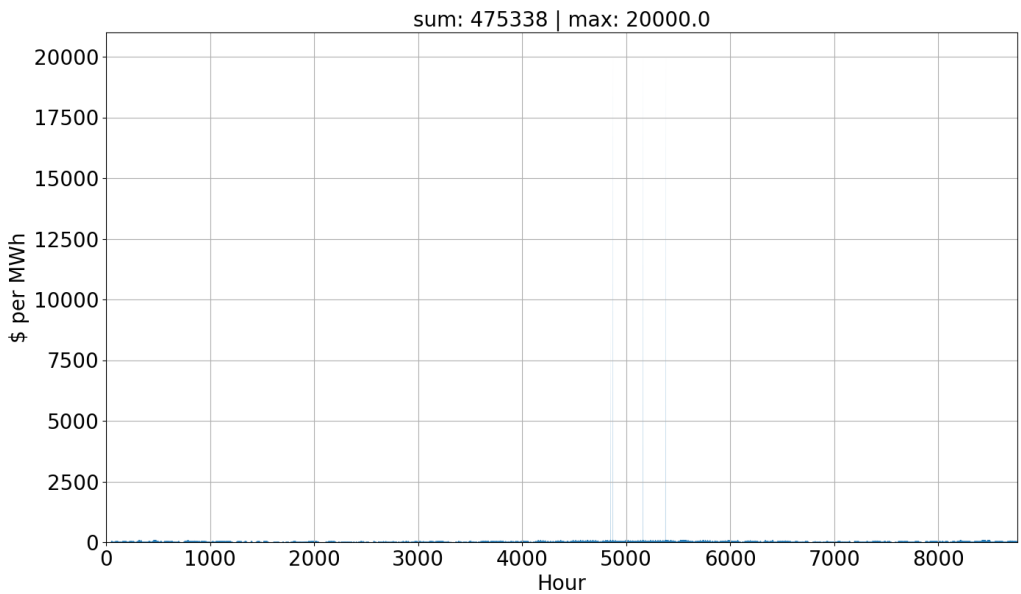


Figure 9: Electricity Price Profile

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